# JaalNet (जालNet)

## — Real-Time UPI Mule-Fraud Defense Platform




Powered by In-Memory Exasol Graph Clustering,
Dual-Tier GNN Inference (**<2ms**),
and Autonomous Regulatory Triage

**Team:** Code Owl

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**Repository:** https://github.com/kumarshivesh465-hue/Jaal-Net.git

# The UPI Mule Fraud Problem & Stakeholders

**14+ Billion** | monthly UPI transactions (~$250B volume)

**Instant, irreversible settlements**
**demand real-time interception**

**PSPs & Banks** SBI, HDFC, ICICI, Axis
compliance fines, reputational risk

**Fintech Aggregators** PhonePe, Google Pay, Paytm
synthetic fraud, high chargebacks

**Law Enforcement & Cyber Crime Units**
investigative delays (4-24 hrs)

**Indian citizens**
victims of digital scams using mule accounts



# JaalNet Solution & Core Value Propositions



## Real-time interception

stop mule syndicates in flight in sub-2 ms

## Dual-tier hybrid AI defense

ONNX GNN for deep detection,
CatBoost for coldstart coverage

## Exasol in-database intelligence

louvain community detection for
live graph insights

## Zero false positives

Merchant Shield protects ;
legitimate commerce

## Automated regulatory triage

instant debit freezes &
FIU-IND STR generation

# Unique Technical Innovations & Differentiators




**In-database parallel Louvain UDF clustering**

100K+ transactions clustered in <500ms

**Dual-model cold-start engine**

routes unseen accounts to CatBoost **(0.9961 AUC-ROC)**

**Autonomous AI Investigator (“War Room”)**

reduces investigation time from 4 hours to ~4 seconds

**Pre-Flight Merchant Shield**

protects high-volume merchants from false freezes

**4D Temporal Velocity Scrubber**

models fund flow for anomaly detection

# System Architecture & Real-Time Data Flow

From UPI stream to **real-time risk action**

### UPI PAYMENT STREAM

Incoming switch transactions ingested <1ms



### EXASOL IN-MEMORY MPP

Live transaction ledger & account graph



### PYTHON UDFS & PYEXASOL POOLING

Parallel execution with connection pooling



### DUAL-TIER AI SERVING

**TIER 1: KNOWN NODES**
TGAT + GraphSAGE
(graph intelligence)

**TIER 2: COLD-START**
CatBoost
(ML for unseen)

### COMMAND CENTER & WAR ROOM

Real-time risk reactor
3D visualization
Automated triage



### REAL-TIME FEEDBACK LOOP

Decisions and signals flow back to stop risk and strengthen the system continuously

# Deep Exasol Technology Utilization

**<0.8 ms**
subgraph extraction within 50ms UPI clearance SLA

**parallel modular Louvain clustering**
inside Exasol memory with no network serialization

**thread-safe PyExasol connection pool**
managing concurrent queries and audit logs

**immutable decision audit ledger**
with model versioning and SHA-256 hashes

**embbedded simulation fallback mode**
for high availability in all environments



# AI/ML Models & Benchmark Metrics

**High accuracy. Low latency. Built for real-time scale.**

## TGAT + GraphSAGE

20K nodes | 100K transactions

structural embeddings for fraud typologies

## CatBoost cold-start

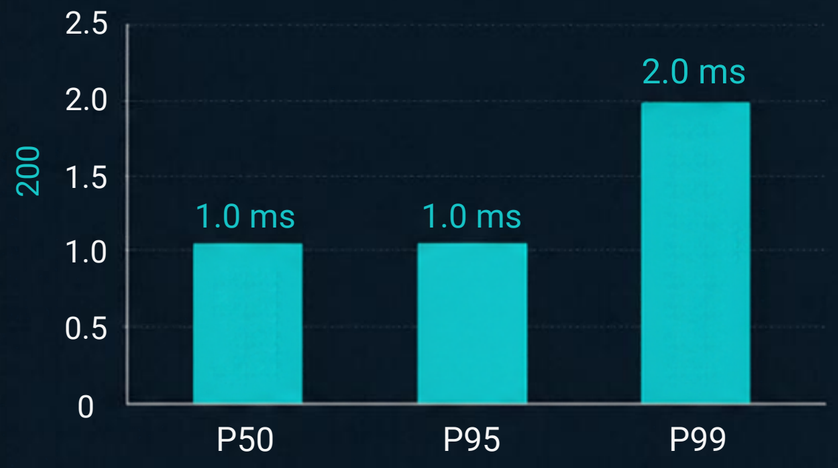
| 0.9961 | 98.4% | 97.2% |
|---|---|---|
| AUC-ROC | Precision | Recall |

## Throughput

**413.34**

requests/second

### Latency Benchmarks (single worker)




well within 50ms SLA

### Precision–Recall Curve




**203**

**tests passed**

**0**

**failures**

unit/integration test suite

# Business Impact & Regulatory Alignment

Compliance by design. Impact by results.




## REGULATORY ALIGNMENT

| | |
|---|---|
| **RBI** | RBI Master Directions & Banking Regulation Act mandates real-time blocking |
| **FIU-IND** | FIU-IND automated electronic STR report generation |
| **PMLA 2002 & CRPC 91** | Autonomous lien placements and audit trails |
| **I4C / CFCFRMS** | Typology tagging integration |

## BUSINESS IMPACT

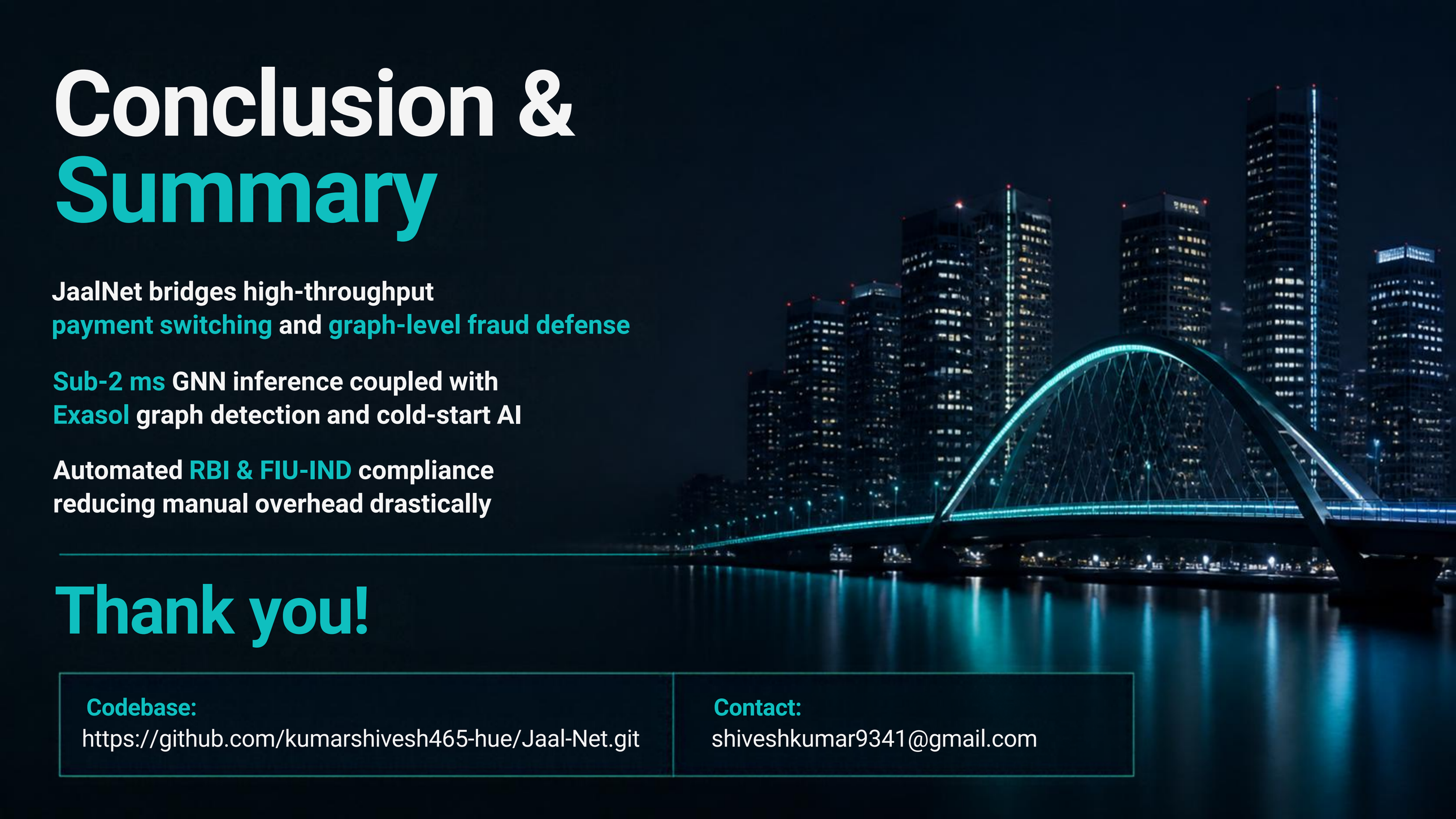
**85%**
**fraud loss reduction**
Protecting revenue at scale

**80%**
**compliance overhead cut**
Lower cost. Higher efficiency.

**Zero**
**false-positive freezes**
No disruption to genuine merchants

JaalNet ensures **compliance**. JaalNet drives **impact.**

# Conclusion & Summary



**JaalNet bridges high-throughput payment switching and graph-level fraud defense**

**Sub-2 ms GNN inference coupled with Exasol graph detection and cold-start AI**

**Automated RBI & FIU-IND compliance reducing manual overhead drastically**

# Thank you!


**Codebase:**
https://github.com/kumarshivesh465-hue/Jaal-Net.git

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